United International University

School of Science and Engineering

Mid Term Exam   Trimester: Spring 2025

Course Title: Coordinate Geometry and Vector Analysis

Course Code: Math 2201   Marks: 30

Total Time: 1 hour and 30 minutes

Answer all questions.

1. Let L_1 and L_2 be the lines [12]

$$L_1: \quad x = 1 + 7t, \quad y = 3 + t, \quad z = 5 - 3t$$
$$L_2: \quad x = 4 - t, \quad y = 6, \quad z = 7 + 2t$$

- **i)** Are the lines parallel?
- **ii)** Do the lines intersect?
- **iii)** Find the distance between lines?
- **iv)** Find the intersection point of the line L_2 and the plane $x - y + z = 1$

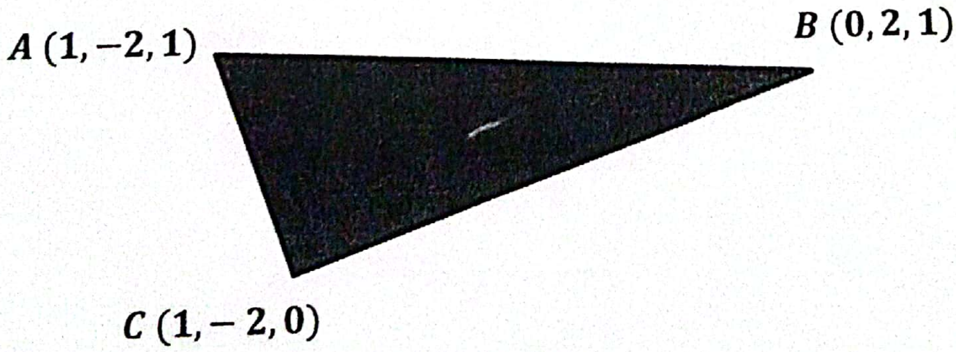
2.

a) Evaluate the double integral $\int_0^{ln5} \int_0^4 2xye^{yx^2}\, dxdy$ [4]

b) Use a double integral to find the volume of region bounded above by the plane $x + y + z = 3$ and below by the rectangle with equations $x = -2$, $y = 1$, $x = 0$ and $y = 5$. [4]

3.

a) [5]



- **i)** Find the area of the triangle ABC.
- **ii)** Find the angle between vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b) Find the projection of $q =< 1, -1,\ 0 >$ along $p =< 0, 0, 1 >$. [2.5]

c) Determine whether the vectors lie in the same plane [2.5]

$p =< 1, 0, 0 >, q =< 2, 0, 1 >, r =< 0, 3, 1 >$.